

2024 HS CS Programming Bootcamp - Part 1

Do not be conformed to this world, but be transformed by the renewal of your mind, that by testing you may discern what is the will of God, what is good and acceptable and perfect. Romans 12:2 (ESV)

Course Description

CS Programming Bootcamp - Part 1 will introduce the students to the fundamentals of programming using the python programming language. The course includes weekly lectures that cover the topics needed to be able to write programs using only code. Each lecture includes multiple lessons with examples and exercises. The course culminates in a large project that the students will develop that will utilize all of the skills learned.

Materials Needed

1. A laptop (mac, windows, linux, chromebook) with internet connectivity.
 - A Mouse is highly recommended.
 - The student will need to verify that their laptop is completely charged or bring the laptop charger.
2. A free / student github.com account.

Course Pre-requisites

1. The student should be able to type at a roughly a 20-25 words per minute pace. This pace is what is considered a beginners pace when starting a typing class for children ages 15-18. If they are considerable slower than this pace, the student should consider practicing several hours each week until the semester begins.

Student Classroom Expectations

1. Students will be attentive and follow along during lectures and will ask relevant and appropriate questions by raising their hand and waiting to be acknowledged by the instructor during lectures.
2. Students will work on the exercises in the time given and if they are not completed will be considered homework.
3. Students will be respectful of all the people in the classroom and will not belittle, disparage, or insult any other individual no matter the circumstances.
4. Student will be in class, with their laptop open, and logged in ready to start by the time class starts. If these conditions are not met the student will be considered tardy (some leeway can be given based on circumstances).
5. The student will not browse the internet at all during lectures and will not be on any unapproved websites during exercises. The only approved websites are directly related to the tasks given. Failure to comply with this will be grounds for discipline as outlined in the student handbook.

Outside Classroom Expectations

The student should expect to spend at least a few hours a week outside of class working, studying and honing their developmental skills. During the final project, most of the work will need to be done outside of the classroom and several hours each week should be expected.

Course Outline

- Week 1
 - Course Intro
 - Github - 1 - codespaces setup
 - Programming 101 - 1 - hello world
- Week 2
 - Programming 101 - 2 - variables
 - Programming 101 - 3 - numbers
 - Programming 101 - 4 - strings
- Week 3
 - Developer Fundamentals - 1 - cli overview
 - Programming 101 - 5 - comparisons conditions and booleans
 - Programming 101 - 6 - user input
- Week 4
 - Developer Fundamentals - 2 - repositories git basics
 - Programming 101 - 7 - else if
 - Programming 101 - 8 - nesting and multiple conditions
- Week 5
 - Programming 101 - 9 - loops
 - Programming 102 - 1 - lists
 - Programming 102 - 2 - iterate through lists
- Week 6
 - Programming 102 - 3 - modifying lists
 - Programming 102 - 4 - lists in lists
- Week 7
 - Programming 102 - 5 - key value pairs
 - Programming 102 - 6 - functions
- Week 8
 - Programming 102 - 7 - function arguments and parameters
 - Programming 102 - 8 - function return values
 - Programming 102 - 9 - function scope
- Week 9
 - Python 101 - 1 - classes
 - Python 101 - 2 - class methods
 - Python 101 - 3 - class inheritance
- Week 10
 - Python 101 - 4 - error handling
 - Python 101 - 5 - standard libraries
- Week 11
 - Python 101 - 6 - splitting code and modules

- Github - 2 - codespaces configuration
 - Python 101 - 7 - graphical libraries
- Week 12 - 14
 - Final Project Workshop